

Creating a Temporally Consistent Spatial Framework for the Analysis of Inter-Regional Migration in Australia

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ABSTRACT

Changes in zonal boundaries severely compromise the reliability of comparisons made from time series data. We examine problems and solutions to the development of a temporally consistent spatial structure in the context of the literature on zone design. Four main approaches to the development of compatible spatial systems are identified and their merits discussed. One of these, the 'designer' solution, is used to establish a consistent spatial structure for the analysis of inter-regional migration in Australia over the period 1976 to 1996. It is shown that intercensal boundary changes over this period create significant spatial discontinuities, and that variations in the digital boundaries from successive censuses hamper development of an automated solution. We set out the tasks involved in the design of geographical zones, establish criteria for the design of migration regions, and describe the key procedures for implementation. Using these principles, we then devise a system of 69 functional regions which for practical purposes are temporally consistent over the 1981, 1986, 1991 and 1996 censuses. We illustrate the utility of the

database by examining selected facets of Australian inter-regional migration.
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INTRODUCTION

Population geographers have long recognised the importance of zone design, but it is only recently, with the development of high-performance computing and geographical information systems (GIS), that real progress has been made in addressing the issues posed by the use of defined zones for the collection and dissemination of socio-economic data. Within this specialised literature, particular attention has focused on the modifiable areal unit problem (MAUP) and on the development of software to optimise the design of aggregate zones to meet specified criteria (Openshaw, 1977; Openshaw and Rao, 1995). Much less consideration has been given to the issue of temporal comparability in zonal boundaries. This oversight is surprising because changes in spatial boundaries significantly prejudice the utility of data collected on a regular, systematic basis.

This paper examines the issues associated with zone design in the context of an especially

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demanding application: the development of a temporally consistent database for the analysis of inter-regional migration in Australia. This database was developed as part of a wider project to compare internal migration in Australia and Britain over the 20-year period 1976 to 1996 in an age-period-cohort framework (Bell *et al.*, 1999). International comparisons of this type can provide valuable new insights into the dynamics of internal migration (see Stillwell *et al.*, forthcoming a, b), but they present numerous conceptual and methodological challenges (Rees *et al.*, forthcoming; Bell *et al.*, forthcoming), including the harmonisation of the spatial frameworks and age-time observation plans. A crucial precursor to the former is the establishment of a temporally consistent set of zones within each country.

We begin by examining a series of general issues associated with the design of geographical zones and identify four possible approaches to the problems of establishing temporally consistent data. Next we outline the nature of the migration data available from the Australian Census and review the problems posed by the boundary changes that have occurred. The following section sets out criteria for the design of migration regions and applies these principles to derive a system of 69 zones which are spatially consistent over the 1981, 1986, 1991 and 1996 censuses. Brief examples are provided to illustrate the utility of the database.

ISSUES IN ZONE DESIGN

Aggregating data to defined zones is a common method of presenting spatial information. For example, the Australian Bureau of Statistics (ABS) publishes Census and other statistics for several hierarchical spatial systems which collectively form the Australian Standard Geographic Classification (ASGC). The main structure of the ASGC consists of five hierarchical levels comprising Census Collection Districts (CDs), Statistical Local Areas (SLAs), Statistical Subdivisions (SSDs), Statistical Divisions (SDs) and States/Territories (Australian Bureau of Statistics, 1996). This structure provides for a range of territorial breakdowns and allows considerable flexibility in reaggregation. In practice, however, the ASGC suffers

a number of limitations.

One problem, common to many spatial systems, is that the zonal structure represents a compromise between several competing objectives. The basic building blocks of the ASGC (CDs), for example, are designed to facilitate data collection rather than to serve as units for data dissemination. Political and administrative boundaries also represent a major constraint on zone design. Thus, SLAs are constructed to coincide with, or aggregate to, legal Local Government Areas (LGAs) which vary widely both in population and in geographical size. Similarly, SDs are designed as subsets of the states and territories. While data output for these geographical divisions may be vital for sound governance, they have little functional basis, bearing no obvious relationship to the underlying distribution of socio-economic variables. Considerable care is therefore needed if the data are to be accorded a meaningful interpretation.

Like all zonal systems the ASGC is also plagued by the modifiable areal unit problem (MAUP). Two aspects of the MAUP are traditionally recognised: those of scale and of zonation (Openshaw and Taylor, 1981; Ebdon, 1985; Fotheringham and Wong, 1991; Wrigley *et al.*, 1996). Scale effects refer to the fact that a country or region may be divided into geographies with differing numbers of spatial units, while zonation effects occur because an area may be divided into the same number of units in a variety of ways. The problem for researchers is that the results of any analysis are directly influenced by these two effects; that is, by the way the territory is divided, and by the level of spatial disaggregation adopted.

In recognition of these issues, considerable attention has been given to establishing the criteria needed to underpin sound zone design (Openshaw, 1978). Cliff *et al.* (1975) and Wise *et al.* (1997) concluded that a good zoning system should be as simple as possible and that the zones should be homogeneous, compact and of equal size.

Substantial progress has been made in automating zone design in accordance with these ideas. Early research by Openshaw (1977) led to development of an automated zone design program (AZP) which was extended during the 1990s to form a Zone Design

System (ZDES), a set of heuristic algorithms that aims to optimise a function of the input data by aggregation of n input zones into m regions, where $n > m$ (Openshaw and Rao, 1995; Openshaw *et al.*, 1998). This work has been used in the UK by Martin (1997, 1998a,b,c) to separate Census output geography from the geography used for data collection by developing 'optimized' census geographies, which seek to standardise population, shape and internal homogeneity of output areas, while preserving all the necessary higher-level statutory boundaries. This methodology forms the basis of the computing system that will be used to define the UK 2001 Census input and output geographies (Office of National Statistics, 1999).

One outstanding problem which is not addressed by these algorithms, and has received much less attention in the literature, is the issue of changes in statistical boundaries, and the difficulties this causes for temporal comparison. With the rapid rise in the amount of information collected on a regular systematic basis, and steady improvements in the facility to store and retrieve these data electronically, the value of time series data is substantially enhanced. Paradoxically, however, successive additions to the time series also increase the risk of boundary changes that prejudice the comparability and hence the utility of this information. If zoning systems are to be designed to maximise the value of the available data in the face of these changing conditions, then a fifth criterion must be added to those outlined above: that of temporal consistency.

Shifting boundaries are an inevitable feature of zonal systems, but the boundaries at the two lowest levels of the ASGC main structure (CDs and SLAs) are especially susceptible to change: the former to maintain appropriate workloads for Census collectors in a context of changing settlement densities; the latter to reflect the seemingly endless litany of local government boundary adjustments and reorganisations. The latter are of particular significance because the consequential shifts in SLA boundaries are echoed at successive levels up the spatial hierarchy, thereby prejudicing temporal comparability throughout the system. Only the states and territories have remained immune

to this ripple effect, although a historical (yet little remarked) change has recently occurred even here, with the creation in mid-1993 of a ninth jurisdiction known as the 'Other Territories'.

APPROACHES TO TEMPORAL CONSISTENCY

There are a range of possible solutions to the problem of establishing time series data on a consistent spatial basis in the face of changing zonal boundaries. At least four distinct approaches can be identified:

- (1) *Freeze history.* One option is to freeze the zone system at a particular point in time and track subsequent splits and amalgamations systematically so that later observations can be adjusted back to the original boundaries. This is the solution adopted by EUROSTAT for monitoring trends at the most detailed (NUTS5) regional level. Its main disadvantage is that the frozen zones become progressively less appropriate with the passage of time. In addition, application of this strategy requires a deliberate decision on the part of the data provider to continue to collect data on the pre-amalgamation boundaries: it is not a solution that can be imposed *post hoc*.
- (2) *Update to contemporary zones.* An alternative strategy is to adopt the most recent geography as the preferred zonal structure and translate the data coverages from previous geographies to this latest spatial system. In practical terms, this requires the creation of look-up tables that link the building blocks at one Census to those at another. The tables can then be used to convert the earlier data to the new geography and the statistics aggregated to larger zones as appropriate for analysis. The enumeration district to postcode directory in the UK operates on precisely this principle. A key advantage of this approach is that converting data at the level of the smallest building blocks provides for maximum flexibility in subsequent aggregation. Its principal shortcoming lies in the potential for inac-

curacies in calculation of the weights used for conversion. As discussed below, interpolations using GIS have a number of potential sources of error but are especially problematic with migration data.

- (3) *Construct designer zones.* A third solution is to construct purpose-built zones from smaller building blocks so as to harmonise the zonal systems from various Censuses on a common set of boundaries. The task then is to find the best fit to optimise specified criteria. While this approach avoids the need for interpolation, it is deceptively complex to apply and may deliver solutions that do not directly match current regional structures.
- (4) *Geo-reference household data.* The difficulties attending the above solutions could be eliminated entirely if individual data were geocoded to discrete addresses. Data could then be assembled on the most appropriate set of boundaries, as defined by the analyst, for the particular problem being investigated, thereby automatically ensuring temporal consistency. The main constraints to this approach are the substantial investment required to establish a geo-referencing system and the concern of many statistical agencies with regard to confidentiality. However, geo-referencing is especially demanding in the case of migration, because geocoding is needed not only for the current place of residence but for the address one or five years previously. Notwithstanding these concerns, procedures of precisely this type will be employed in the 2001 Census of the UK (Office of National Statistics, 1999) but for the time being, at least, they appear a more remote prospect in Australia.

In practice, the choice of approaches will largely be dictated by the way in which the data are collected. Options 1 and 4, for example, are only feasible if the requisite spatial structures have been put in place. However, the nature of the data to be examined is also an important consideration in the choice of method. GIS overlays, which are integral to option 2, are widely used to translate data from one geographical coverage to another. The estimation procedure may be based on a variety

of techniques ranging from simple areal interpolation (Goodchild and Lam, 1980) to more sophisticated dasymetric analysis (Langford *et al.*, 1991). While these procedures are generally satisfactory as a means of estimating aggregate variables, such as total population or total households, they are less reliable for individual characteristics like age or employment status, which may vary systematically across space in ways that are unrelated to the overall distribution of population. Spatial interpolation is especially perilous in the case of migration data because it has been shown that the likelihood of inter-zonal migration rises with proximity to zonal boundaries (Rogerson, 1990; Boyle and Flowerdew, 1997), and hence depends on nuances in the distribution of population within a zone which are not readily captured by overlay techniques.

Option 3 eliminates the need for interpolation by aggregating data from smaller building blocks to a common zonal structure. This is a potentially expensive solution because it will generally require reprocessing of each census to generate special tabulations on the common spatial boundaries. In practice, however, such reprocessing may hold a number of advantages. In Australia, for example, the extent of cross-classification made available in Census output depends on the size of the table population, and the Australian Bureau of Statistics (ABS) further seeks to protect individual confidentiality by subjecting small cell counts to randomisation. Reprocessing Census data to generate customised tabulations on designer zones not only avoids compounding errors from the aggregation of small cell counts, but may be the only reliable method of acquiring the full range of population characteristics actually required for the analysis. Such considerations loom particularly large in the context of migration because detailed origin-destination matrices rapidly approach the size limits for data release set by ABS.

THE AUSTRALIAN CONTEXT

Data on Internal Migration

In Australia the only comprehensive source of information on internal migration derives from the quinquennial Census of Population

Table 1. Summary of boundary changes between statistical division, states and territories, 1976 to 1996.

State or territory	Number of SDs ¹		Number affected by boundary changes				Unchanged 1976–96
	1976	1996	1976–81	1981–86	1986–91	1991–96	
New South Wales	12	12	4	5	0	5	6
Victoria	12	11	4	5	7	11	0
Queensland	11	11	2	0	2	2	8
South Australia	7	7	0	2	2	1	3
Western Australia	9	9	0	0	3	5	3
Tasmania	4	4	0	0	0	4	0
Northern Territory	2	2	0	2	2	0	0
ACT	2	2	0	0	0	1	1
Other Territories	0	1	0	0	0	1	0
Total Australia	59	59	10	14	16	30	21

¹ Excludes the SDs designated 'Offshore and Migratory' in the six states and the NT.

Source: Australian Bureau of Statistics, 1992, 1996, and Census Publications (various years).

(ACT = Australian Capital Territory; NT = Northern Territory.)

and Housing. The Census collects migration data through a series of multi-part questions that seek each person's place of usual residence on Census night and their usual address five years previously. Similar data have been collected at each Census since 1971. Since 1976 a question on place of usual residence one year ago has also been included.

At the 1971 Census, place of previous residence was only coded to a relatively coarse level, distinguishing between urban and rural residence within 68 statistical divisions (SDs) of Australia (Rowland, 1979: 33). The number of SDs has subsequently been reduced to 59. Since 1976, however, usual residence has been coded to SLA level (known as Census LGAs prior to 1986), which represents a much finer level of spatial disaggregation. At the 1996 Census, for example, Australia was divided into more than 1330 SLAs, thereby potentially providing an origin–destination migration matrix of more than 1.8 million cells.

Together, these data represent an invaluable source of information on the changing patterns of population movement within Australia. Coupling the five-year migration questions asked at the 1971, 1976, 1981, 1986, 1991 and 1996 Censuses can potentially provide an uninterrupted sequence of intercensal migration flows spanning more than a quarter of a century. Linking the spatial information with data on other population characteristics col-

lected at the Census promises a rich, multi-dimensional picture of the dynamics of population change.

From a demographic perspective, only three minor deficiencies mar the utility of these data. The first was the decision to process only 50% of returns from the 1976 Census, thereby introducing sampling error into the 1971–76 migration data. The second occurred when, in another effort to save money, the one-year migration question at the 1991 Census only sought information on place of residence at state or territory level, rather than for SLAs. The third was the discontinuity brought about by the decision to conduct the 1991 and 1996 Censuses on 6 August instead of the traditional date of 30 June. Since the migration questions specifically ask for information on place of usual residence exactly five years prior to Census day, there is now a small discontinuity in the series which leaves uncovered the interval between 30 June and 6 August 1986. More serious than these three deficiencies from a practical perspective are the limitations arising from the myriad changes that have occurred in the boundaries of SLAs at successive Censuses.

Changes in Statistical Boundaries

Since SLAs are the smallest spatial units to which usual residence is coded, any changes to

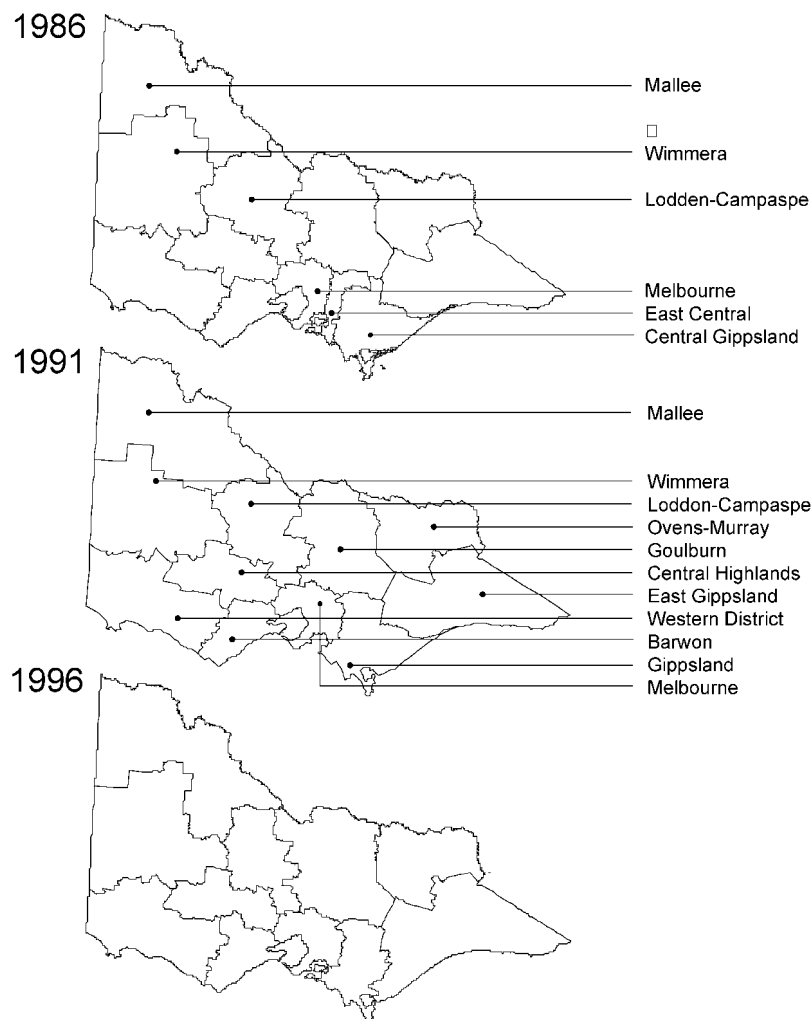


Figure 1. Statistical Divisions of Victoria, 1986, 1991 and 1996. Note: Labelled SDs changed their boundaries in the following five-year interval.

these boundaries directly prejudice the temporal comparability of migration data. Table 1 provides a concise summary of the incidence of boundary changes over the four intercensal periods between 1976 and 1996 at the next level but one in the ASGC main hierarchy: Statistical Divisions. The changes at this scale are of particular interest because SDs were originally designed to delimit areas which would remain stable for statistical purposes over a period of 20 years or more (Australian Bureau of Statistics, 1996: 15). In practice, however, it is apparent that considerable change has taken place. In 1996 there were 59 SDs in Australia (excluding the seven desig-

nated to cover offshore and migratory populations in the six states and the Northern Territory). This number was unaltered from 1976, although two compensating changes did in fact occur over the intervening period: the carving-up of Victoria's East Central SD in January 1991, and the creation of the new 'Other Territories' SD in mid-1993. Of the 58 SDs that survived intact, however, only 21 remained unchanged throughout the entire 20-year interval.

In some instances these changes involved only small shifts in population or area. For example, the five changes recorded in New South Wales between 1981 and 1986 were the

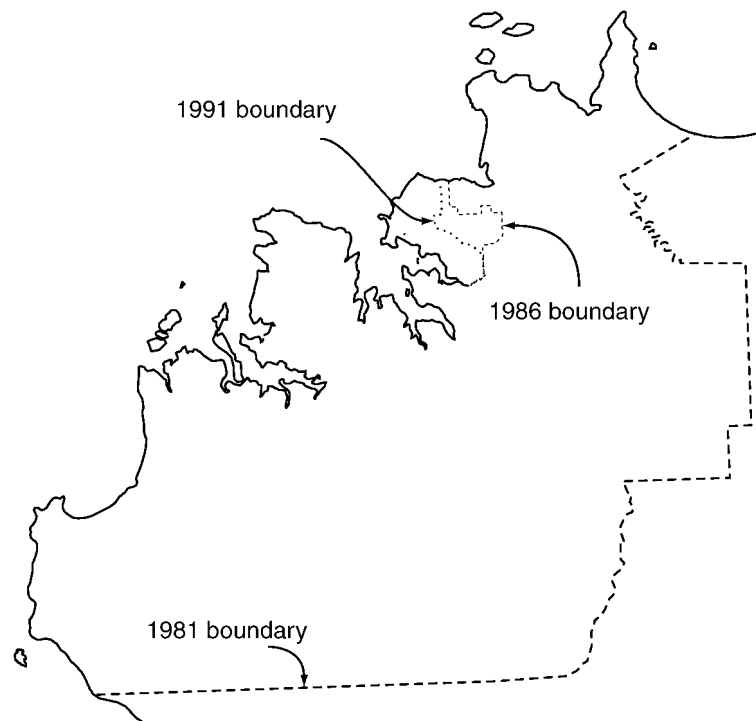


Figure 2. Boundaries of the Darwin (Northern Territory) Statistical Division, 1981, 1986 and 1991.

product of just three territorial transfers between SDs, each of which involved only a little over 100 hectares of land. The 1991–96 changes appear to have been of a similar magnitude. In Victoria, on the other hand, there has been a sequence of major boundary revisions in successive intercensal periods since 1981, culminating in changes to all 11 SDs between 1991 and 1996 in the wake of a massive restructuring of local government. The magnitude of these changes between 1986, 1991 and 1996 is readily apparent from Fig. 1, in which the labelled SDs indicate where changes occurred over the ensuing five years.

Not surprisingly, the boundaries demarcating the capital city SDs in each state have proven the most fragile: Melbourne, Brisbane, Adelaide, Hobart and Darwin all underwent boundary adjustments on at least one occasion over the 20-year period. Of the eight capitals, only the Sydney, Perth and Canberra SDs remained intact. These changes generally involved expansion of the SD boundary to accommodate unforeseen population growth and were effected either by annexation of

surrounding shires or through the relocation of existing SLA boundaries. Darwin offers a unique case in which the SD underwent successive contractions, rather than expansions. Between 1981 and 1986 the Darwin SD was reduced by 97% from an area of 12,920 km² to just 337 km², and in the following five years it was cut by a further third to just 232 km² (Fig. 2).

In the non-metropolitan parts of the country, SD boundaries have been much more resilient. In Queensland and South Australia, for example, only one boundary relocation was recorded outside the capital cities between 1976 and 1996. Western Australia, with four such changes, has been more severely affected, but it is notable that these shifts were confined entirely to the 1991–96 intercensal period, as were those in Tasmania. With the passage of time it was inevitable that the original configuration of SD boundaries would become progressively more susceptible to change, either to accommodate unforeseen population growth (especially on the metropolitan fringe) or as a result of local government reorganisations.

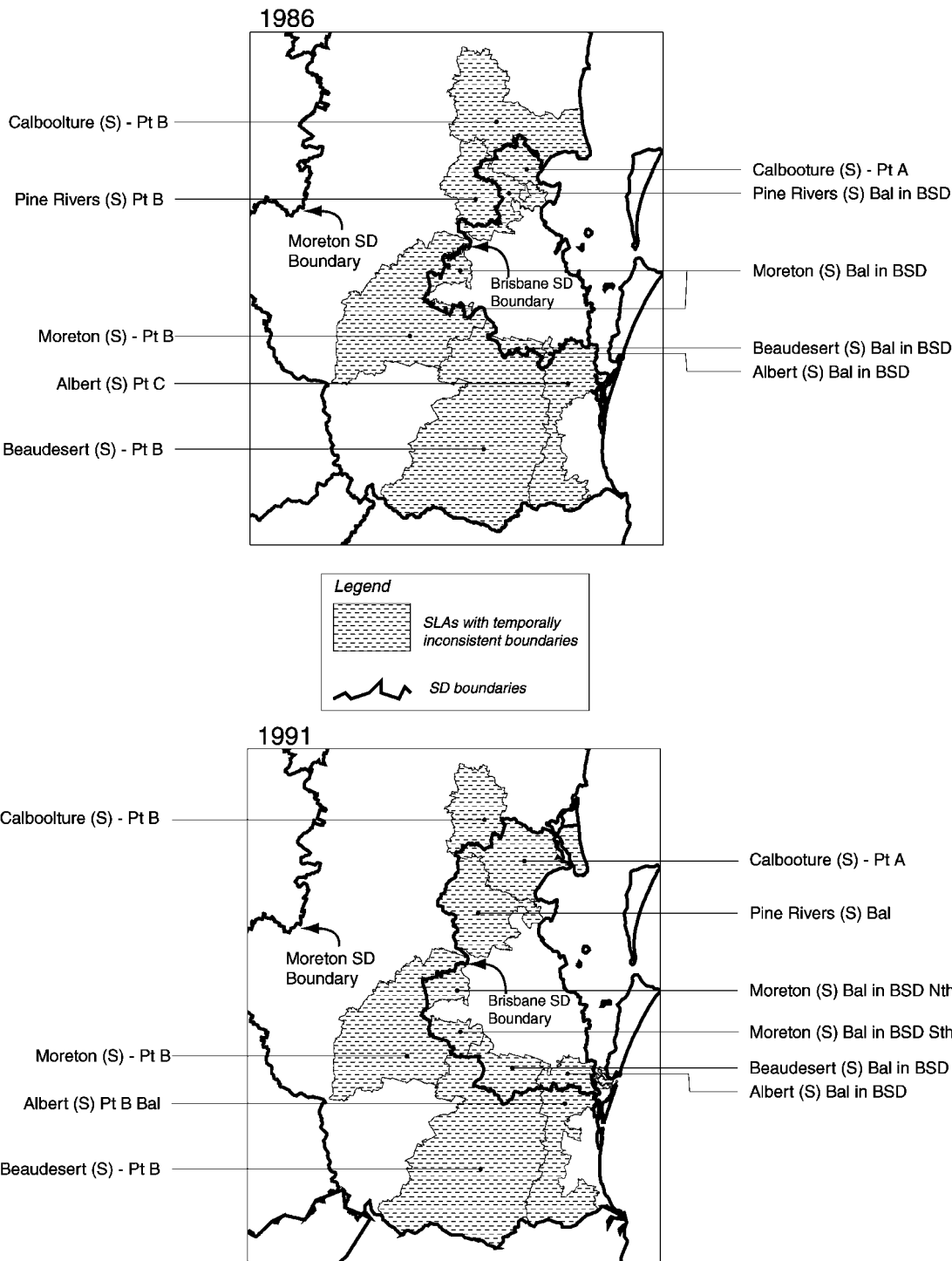


Figure 3. Boundaries of the Brisbane (Queensland) Statistical Division, 1986 and 1991.

Intercensal boundary changes for 1986–91 and 1991–96 are summarised in Census editions of the ASGC, and similar information for earlier intervals can be found in regular

Census outputs (see, for example, Australian Bureau of Statistics, 1978, 1983, 1988). Although the ABS no longer publishes estimates of the population involved, two key

types of changes can be distinguished:

- those involving the transfer of one or more complete SLAs from one SD to another
- those arising from the relocation of an SLA boundary.

Because usual residence is only coded to SLA level, these differences are crucial in the construction of temporally consistent migration regions. Unfortunately, most of the changes that have occurred over the 20-year time frame at the SD level have involved SLA boundary relocations. Of the 70 instances identified in Table 1 in which SDs were affected by boundary changes, only in 13 cases was the movement confined to complete SLAs. Examples include the transfer of Queensland's Noosa Shire from Wide Bay-Burnett to Moreton (1976–81), Boddington Shire from Upper Great Southern to the South West of WA (1991–96), and in South Australia the 1981–86 shift in the Adelaide SD boundary to incorporate the whole of Willunga District Council (DC), but to excise those parts of Onkaparinga DC and Gumeracha DC previously included.

The remaining 57 cases involved the relocation of SLA boundaries and hence the transfer of part-SLAs from one SD to another. These in turn can be divided into several groups depending on their complexity. In a few instances the changes could be classed as *trivial*. The boundary alteration recorded for Perth SD between 1991 and 1996, for example, represented a minor annexation of areas of water, while over the same interval Adelaide SD gained a small area of previously unincorporated land (Australian Bureau of Statistics, 1996). Few people were involved in either case.

More common were transfers that involved a *simple boundary realignment*. These sometimes occurred in response to local government boundary changes, but in urban areas they often represented a statistical rather than an administrative decision, involving the relocation of an SLA boundary positioned to segment the urbanising parts of an LGA from its predominantly rural hinterland. These realignments involved considerable variation in geographical area. As noted earlier, several of the changes in New South Wales involved

only around 100 ha of land. In contrast, Victoria's Goulburn SD gained more than 28,000 ha from the East Central SD between 1981 and 1986, while in Western Australia the following intercensal period saw the transfer of 388,500 ha from the Pilbara to the Central SD following a boundary adjustment in favour of the Shire of Exmouth.

In many cases *multiple realignments* occurred, involving gains from one or more boundary revisions, sometimes offset by losses elsewhere. Changes of this type disrupted continuity in the boundary between Brisbane and Moreton in southeast Queensland between 1986 and 1991. As Fig. 3 shows, these changes were solely a function of shifts in SLA boundaries designed to accommodate urban expansion, but with no accompanying change in the boundaries of Legal Local Government Areas. Thus, Brisbane SD expanded simply by relocating the SLA boundaries that divided Caboolture, Pine Rivers, Albert, Beaudesert and Moreton to encompass larger proportions of those Shires.

At the extreme, such changes merge into *major boundary restructures* like those that occurred in both Victoria and Tasmania in the early 1990s, which resulted in the wholesale redefinition of local government boundaries. These often complex changes pose a formidable challenge to the definition of temporally consistent regions.

DEFINING TEMPORALLY CONSISTENT MIGRATION REGIONS

The design of geographical zones involves three principal tasks:

- defining the goals the zones must satisfy
- assembling the requisite data into suitable database and GIS systems
- applying appropriate procedures to construct the new system of zones.

Defining the Goals

For the present purposes of comparing migration over time and between nations, the key criterion was that the zones should be relatively large, functional regions for which a consistent temporal boundary could be de-

finer covering the 1976 to 1996 interval. Since migration data are only coded to SLA level, a fundamental constraint was that the zones should comprise one or more whole SLAs with a common outer boundary.

Of the zonal systems defined in the ASGC, SDs most closely match this criterion. SDs were originally designed in the mid-1960s and were intended to form '... large ... relatively homogeneous regions characterised by identifiable social and economic links between the inhabitants and between the economic units within the region, under the unifying influence of one or more major towns or cities' (Australian Bureau of Statistics, 1992). Apart from the boundary changes discussed above, the principal limitation of SDs for migration analysis is that the capital city in each state and territory is represented by a single zone. Because of the high degree of metropolitan primacy in Australia, the five capital city SDs together house almost 60% of the Australian population. Leaving these as five single zones would therefore mask a considerable proportion of the total population movement occurring within the nation. It would also obscure the key processes of suburbanisation and urban regeneration that have been crucial elements in the development of Australian cities over the past two decades. While the ASGC provides two alternative zonal systems which are intermediate in size between SLAs and SDs (SSDs and Statistical Regions), these do not capture the underlying morphology of the cities in a way that reflects these currents of population movement. For migration analysis, a better zonal system is one based on concentric rings that differentiate the inner, middle and outer suburbs, such as have been assembled by the planning agencies in several mainland states (New South Wales Department of Urban Affairs and Planning, 1995; Victoria Department of Infrastructure, 1996; Western Australia Ministry for Planning, 1997; Queensland Department of Communication and Information Local Government and Planning, 1999).

Seeking Temporal Consistency in a GIS Framework

GIS offers the obvious framework in which to search for temporally consistent zonal bound-

aries. Digital overlays of the spatial systems employed at successive Censuses should provide direct and unambiguous evidence of boundary changes and enable temporally consistent regions to be identified. To these ends, digitised boundaries for SLAs and SDs at the 1981, 1986, 1991 and 1996 Censuses were acquired from the Australian Bureau of Statistics, and a spatial database constructed. CD boundaries and their associated populations at each Census were also assembled to provide a basis for estimating the size of the population involved in any boundary shifts. Although the migration database aimed to cover the four five-year periods since 1976, it should be noted that specific boundaries for 1976 were not needed because the usual residence at each point in time is coded to SLAs as defined at the time the Census was taken, rather than to the geography that existed one or five years previously. Thus, usual residence in 1976, as recorded in the 1981 Census, was coded to 1981 Census LGAs, rather than to 1976 boundaries.

In principle, the task of defining temporally consistent regions seems readily suited to automation within a GIS environment using procedures similar to those encapsulated in zone design algorithms such as ZDES (Openshaw and Rao, 1995). Thus one might envisage an algorithm that searched heuristically for a temporally consistent set of regions by progressively aggregating areas adjoining a specified set of seed zones, such as SLAs, subject to various constraints and optimisation criteria. Alternatively, one might seek to dissolve any non-conforming boundaries and then partition the territory based on the temporally consistent zones that remain.

In practice, automation is severely hampered by inconsistencies in digital boundaries. Because the geographical zones employed by the ABS have been redigitised at each Census, discrepancies often occur even where no territorial change has actually taken place. Indeed, it is unusual to find instances in which the boundaries from different Censuses directly coincide. Such variations may reflect differences in digitising technique and accuracy, map projection or subsequent transformation, but they can also be an artefact of the resolution at which the digitised boundaries

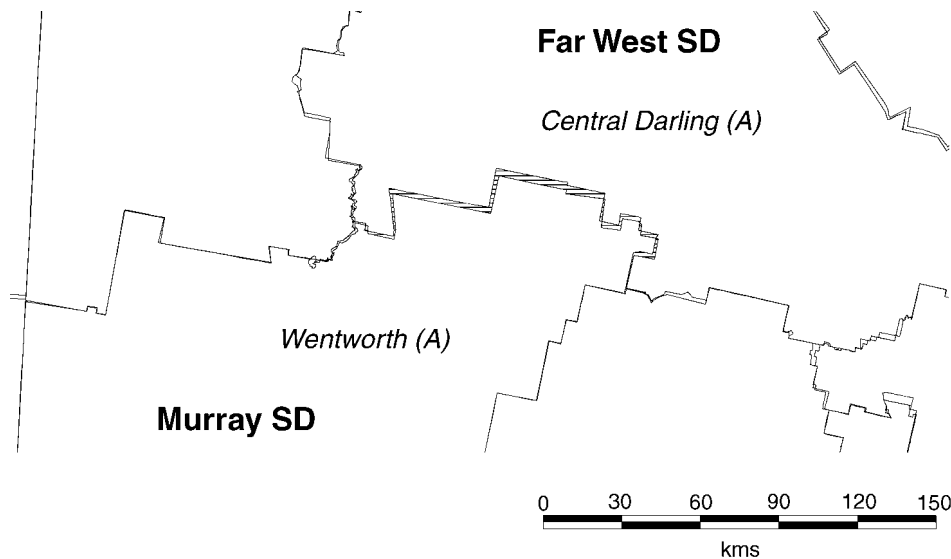


Figure 4. Anomalies in digital boundaries between the Murray and Far West SDs of New South Wales, 1981 and 1996.

are made available. The digital SD boundaries provided with the 1996 Census product CDA-TA96, for example, are an abbreviated set which do not directly match the more detailed digital boundaries provided for SLAs on the same CD-ROM.

The extent of these discrepancies in the digital boundaries varies, but can be surprisingly large, as illustrated in Fig. 4. According to the digital data these boundaries at the intersection of the Murray and the Far West SDs underwent a substantial change between

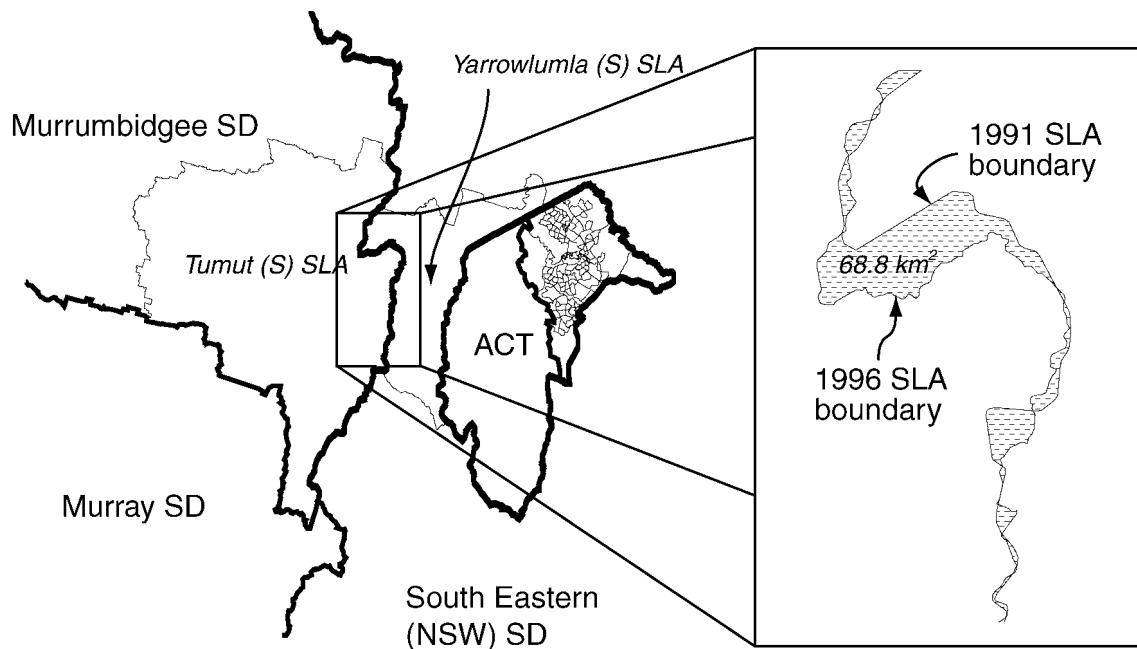


Figure 5. Anomalies in digital boundaries between the Tumut and Yarrowlumla Shires of New South Wales, 1991 and 1996.

1981 and 1996. In fact, no such change took place. Yet the maps indicate a transfer of more than 340 km² from Wentworth to Central Darling. This discrepancy in the digital boundaries is substantially larger than many of the intercensal changes identified by the ABS in the ASGC. Indeed, many of the smaller boundary adjustments identified by the ABS in the published record (the three 1981–86 boundary adjustments of around 100 ha between SDs of New South Wales mentioned earlier, for example) are simply not evident from comparison of the digital boundary data. Figure 5 provides a rather different example of a boundary discrepancy, in this case reflecting a sequence of digitising differences along a largely featureless boundary between the Shires of Tumut and Yarrolumla to the west of the national capital, Canberra. The gap of more than 68.8 km² generated by these variations in digitising is equivalent to the combined areas of the central city LGAs of Sydney, Melbourne, Adelaide and Perth (69.9 km²). In this situation, automation is clearly prejudiced by the difficulties of distinguishing real changes in zone boundaries from digitising errors.

Constructing Temporal Statistical Divisions (TSDs)

In the absence of a reliable, automated procedure for zone design, a manual approach was employed to develop a set of temporally consistent regions based around a combination of SDs and the concentric zones identified by planning agencies in the five mainland state capitals. GIS spatial overlays together with published ABS documentation were used to compare the SD and SLA boundaries at each of the four Censuses from 1981 to 1996, to identify where boundary changes between SDs had occurred. A heuristic procedure was then used to search for the nearest set of temporally consistent boundaries at SLA level. The overall approach, then, falls into the class of *designer solutions* as outlined earlier.

Four main principles were employed to guide the search and adjustment procedure:

- Where possible, 1996 SDs were adopted as the standard, and earlier boundaries were

adjusted to match. This was possible only where the change was confined to the transfer between SDs of one or more complete, pre-existing SLAs.

- SD boundaries for earlier Censuses were adopted when the 1996 boundary did not provide a viable option. This could occur, for example, if the 1996 SD boundary was shifted to encompass a newly created SLA.
- Where none of the SD boundaries provided consistency, the nearest temporally coincident SLA boundaries were adopted. If more than one option was available, the choice was made so as to minimise the aggregate deviation, in terms of total population, from the 1996 SD boundary.
- Where minor realignments had been made to SD boundaries involving part SLAs with comparatively small geographical areas, the size of the population involved was estimated from CD data. If the population in the realigned area represented less than 1% of the aggregate population of any of the affected SDs, the inconsistency was accepted: that is, no realignment was made. The same principles were applied to define temporally consistent zones within the mainland state capitals.

The net result of these procedures is a system of 69 functional regions, defined collectively as Temporal Statistical Divisions or TSDs, which cover the whole of Australia, excluding the newly-created Other Territories, without gaps or overlaps. The zones are depicted in Figs 6 and 7 and are defined formally by a set of four look-up tables, one for each Census, that list each of the SLA or Census LGA codes and the code of the TSD to which they were assigned. The TSDs themselves are numbered incrementally within states, using a three-digit identifier in which the first digit represents the state or territory, proceeding from 1 to 8 in the conventional order (see Australian Bureau of Statistics, 1996). To facilitate identification we have generally retained the original SD names for the new TSDs, although in many cases the boundaries are significantly changed.¹

Full comparability is out of reach at this level of spatial disaggregation, and some temporal inconsistencies remain. Nevertheless, the TSD



Figure 6. Temporal Statistical Divisions for Australia.

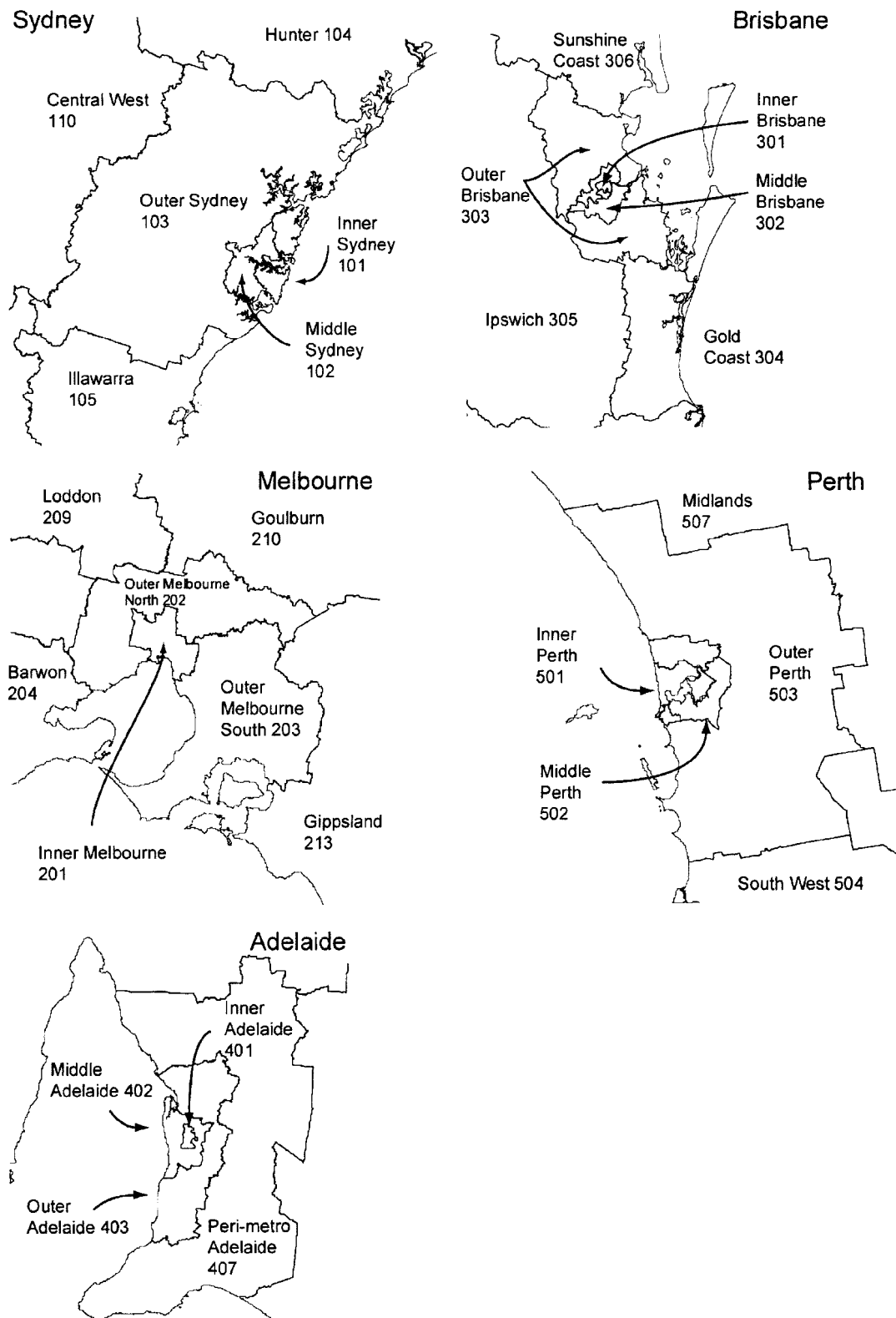


Figure 7. Temporal Statistical Divisions in the mainland state capital cities of Australia.

Table 2. Residual spatial discontinuities between TSDs.

State	Interval	Losing TSD	Gaining TSD	Area (km ²)	Population
VIC	1991-96	Western District	Wimmera	1089	365
VIC	1991-96	Wimmera	Western District	151	20
VIC	1991-96	Central Highlands	Wimmera	20	80
VIC	1991-96	Wimmera	Central Highlands	31	20
VIC	1991-96	Outer Melbourne Nth	Central Highlands	40	180
VIC	1991-96	Inner Melbourne	Outer Melbourne Nth	17	170
VIC	1991-96	Outer Melbourne Nth	Goulbourn	54	1400
VIC	1991-96	Outer Melbourne Sth	Gippsland	31	250
VIC	1991-96	East Gippsland	Ovens-Murray	104	200
SA	1986-91	Outer Adelaide	Inner Adelaide	7	150
WA	1986-91	Pilbara	Central	3895	200
WA	1986-91	South Eastern	Central	1005	50
TAS	1991-96	Mersey-Lyell	Southern	175	50
TAS	1991-96	Northern	Southern	770	0
NT	1981-86	NT - Balance	Darwin	43	2000

system does provide a markedly better temporal match than is afforded by the original SD boundaries, while at the same time preserving the general integrity of the SD geography and extending its utility by dissecting the larger cities. Overall, 51 of the 69 TSDs are spatially consistent across the 1981, 1986, 1991 and 1996 Censuses. In the remaining 18 cases, either no temporally coincident boundary could be found, or the discrepancy was considered too small to be of practical significance. The boundary changes giving rise to these residual discrepancies are set out in Table 2, and it is evident that while they vary widely in area, the populations involved in these changes are generally small. The most significant discontinuity remaining is between Darwin and the NT Balance, reflecting a part of the boundary shift between 1981 and 1986 for which no further adjustment is possible. The only other discrepancy involving more than 1000 people is between Outer Melbourne North and Goulbourn, and in this case the anomaly represents less than 0.8% of the 1996 population of the Goulbourn TSD.

The main features of the TSD structure in each state and territory can be summarised as follows:

- In New South Wales, changes in SD boundaries involved very small transfers of land in

sparsely populated rural areas. As a result, no changes were considered necessary to maintain effective continuity and the TSD boundaries were therefore defined to follow the SD boundaries at successive Censuses. Within Sydney, the Inner, Middle and Outer TSDs are based on the definitions employed by the New South Wales Department of Urban Affairs and Planning (1995).

- Victoria presents a very different situation, with the myriad changes in LGA and SD boundaries since 1981 presenting a formidable challenge. Only in three TSDs—Mallee, Loddon and Barwon—was it possible to establish boundaries which were fully consistent over time, although as Table 2 shows, the temporal inconsistencies that remain elsewhere are quite small. Similarly, the extent of the boundary revisions within Melbourne was such that it proved impossible to identify a system of concentric rings, and the final TSD structure was a compromise designed to reflect another important dimension of the city's social geography by differentiating the inner from the middle and outer suburbs and, within the latter, the more prosperous southern and eastern from the less well-heeled northern and western suburbs.
- There have been few boundary changes in non-metropolitan Queensland, so TSD

boundaries could therefore be defined to match 1996 SDs. In the southeast of the state, however, there was no temporally consistent boundary between Brisbane and Moreton, and an alternative approach was needed. The solution adopted (Fig. 7) was to couple the LGA of Brisbane City with its immediate neighbours (Logan City and the Shires of Pine Rivers and Redland), to define a capital city region which was then split into three concentric rings using SLA boundaries (see Queensland Department of Communication and Information Local Government and Planning, 1999). The remaining parts of the Brisbane SD were then grouped with adjacent SLAs from the surrounding SD of Moreton, and divided radially into three sectors. While the dispersed settlement pattern of southeast Queensland makes it difficult to establish any clear metropolitan boundary, this TSD structure does have the merit of segregating the rapidly urbanising Gold and Sunshine Coasts from the predominantly rural remainder of the Moreton SD.

- Boundary changes between SDs in South Australia over the 1976–1996 period were confined solely to transfers of entire SLAs. TSDs could therefore be defined to match existing SDs. Adelaide itself was further divided into three TSDs in the form of concentric zones.
- Western Australia presents a more complex problem, with two major boundary realignments, one LGA transfer between SDs and the creation of one new LGA during the study period. The changes involving whole LGAs are readily dealt with, but the realignments proved less tractable and their legacy is a discontinuity in the boundaries of the Pilbara, South Eastern and Central TSDs. Within Perth, as in other cities, three additional TSDs were created in the form of concentric zones, based on the boundaries defined by the Western Australian Ministry for Planning (1997).
- In Tasmania, a temporally consistent region for Hobart was created by expanding the 1996 Hobart SD to encompass four SLAs from the adjoining Southern SD (Southern Midlands, Tasman, Sorrell (Pt B) and Glamorgan/Spring Bay). Boundary changes

Table 3. Population sizes of SDs and TSDs, 1996.

	TSDs	SDs
Minimum	12,205	301
Maximum	1,750,728	3,717,508
Range	1,738,523	3,717,207
Mean	256,432	305,083
Standard deviation	349,350	660,468

between other SDs in the 1991–96 interval involved relatively small populations, so the discrepancies were accepted and TSDs were defined to follow the existing SD boundaries.

- In the Northern Territory the 1986 SD boundary provided the most reliable foundation from which to define temporally consistent regions, since this boundary was maintained in the division between parts A and B of Litchfield Shire at both the 1991 and 1996 Censuses. Full temporal consistency could not be extended back to 1981, but the residual discrepancy is small.
- In the ASGC the Australian Capital Territory (ACT) is divided into two SDs, but the population in the Balance region outside Canberra is very small. For current purposes the ACT was therefore defined as a single TSD.

Table 3 summarises the differences between the SD and TSD zonal systems in terms of their 1996 populations. The most significant variation is the much smaller maximum population in the TSD system, resulting from division of the five mainland state capital city SDs. The SD system is dominated overwhelmingly by the two largest cities, Sydney (3.72 million) and Melbourne (3.16 million), which together account for 39% of the national population. In the TSD system, Outer Melbourne South (1.75 million) emerges as the most populous zone, followed by Outer Sydney (1.63 million), Inner Sydney (1.42 million) and Inner Melbourne (1.09 million). All other TSDs house populations of less than a million. While the size distribution is still highly skewed, the TSD system reduces the dominance of Sydney and Melbourne and radically reduces the range from the least to the most populous zones. The creation of these additional zones also signifi-

Table 4. Net internal migration by TSD, New South Wales and Queensland, 1976–81 to 1991–96.

TSD	1976–81	1981–86	1986–91	1991–96
New South Wales				
Inner Sydney	–58,396	–30,227	–44,593	–16,104
Middle Sydney	–68,880	–51,269	–79,115	–60,880
Outer Sydney	65,406	17,692	–12,921	13,149
Hunter	12,592	4,695	9,220	6,735
Illawarra	4,471	–1,479	8,581	5,646
Richmond-Tweed	14,264	9,894	15,916	12,767
Mid-North Coast	20,294	17,483	19,126	11,285
Northern	–4,142	–10,009	–4,358	–12,459
North Western	–1,160	–4,364	–1,829	–5,782
Central West	–4,839	–4,816	–2,470	–4,327
South Eastern	1,146	4,477	5,928	1,431
Murrumbidgee	–3,418	–8,214	–3,745	–4,583
Murray	–156	–560	–513	–1,979
Far West	–2,136	–3,434	–2,204	–3,002
Queensland				
Inner Brisbane	–3,531	–6,583	–320	–152
Middle Brisbane	–18,665	–14,234	–3,271	–1,151
Outer Brisbane	43,211	26,003	27,596	32,509
Gold Coast	25,128	34,224	36,073	40,915
Ipswich	6,489	11,344	9,222	9,544
Sunshine Coast	29,252	23,769	42,743	49,677
Wide Bay-Burnett	1,696	5,892	13,951	14,053
Darling Downs	–1,980	1,756	2,678	–1,212
South West	–2,585	–1,501	–6,237	–3,907
Fitzroy	4,835	1,809	–141	78
Central West	–1,350	–1,674	–2,223	–1,696
Mackay	3,918	3,326	–561	939
Northern	–1,171	3,387	1,527	1,514
Far North	2,837	5,054	7,058	7,202
North West	–5,584	–5,445	–5,721	–6,353

cantly raises the volume of inter-regional migration that is captured by the system, from less than 1.8 million inter-SD migrants to more than 2.6 million moving between TSDs.

ANALYSING TRENDS IN INTER-REGIONAL MIGRATION

The development of a temporally consistent set of regions within Australia represents only the first step in a sequence needed to establish internationally comparable databases of internal migration. Elsewhere, the authors together with colleagues have set out procedures to harmonise the age–time plans of migration data from the Australian Census with British

data for the corresponding 20-year period drawn from the National Health Service Central Register (Bell *et al.*, 1999). We have also proposed a system of city-regions for each country which, it is argued, captures the underlying commonalities in the space economies of two countries with otherwise markedly different physical and settlement geographies (Blake *et al.*, forthcoming). Using these databases we have begun to explore key dimensions of the migration systems in the two countries in a systematic way (see, for example, Stillwell *et al.*, forthcoming a, b).

Even without these additional processes of harmonisation, the TSD migration database represents a powerful resource. By way of

example, Table 4 sets out net internal migration for TSDs in the states of New South Wales and Queensland over the four intercensal periods between 1976–81 and 1991–96. The results reveal marked fluctuations in the magnitude, and even the direction, of net migration. In New South Wales there is evidence of an abrupt slowdown in net out-migration from inner Sydney in the first half of the 1980s, with a corresponding reduction in net migration gains on the fringe. The trend to inner urban renewal is swamped by a massive rise in losses from all three zones between 1986 and 1991, but resumes at an intensified rate in the 1991–96 interval. Brisbane displays a more regular pattern, but in this case the reductions in out-migration extend to the middle suburbs.

Beyond the cities, Table 4 highlights the sharp divide between inland and coastal Australia: the former experiencing sustained out-migration, the latter absorbing rapid growth. But the time series data also reveal distinct variations within these two broad regions: on the coast, migration gains appear increasingly concentrated in a few selected areas, especially the Gold and Sunshine Coasts, Wide Bay Burnett and Far North Queensland, while regions like Mackay, Fitzroy and Northern Queensland, together with much of New South Wales, have begun to languish or lose their attraction. Regions of the interior, on the other hand, especially in New South Wales, display much greater oscillations over the 20-year period, with the magnitude of the migration loss readily traced to a succession of rural crises ranging from periodic drought to cyclic variations in commodity prices and the terms of trade.

CONCLUSION

Censuses represent a substantial investment of public money. Their main purpose is to provide information on the size, composition and distribution of the population that is both comprehensive and up-to-date. Comparison with similar data collected at previous enumerations has the potential to enhance significantly the utility of this resource by providing data on historical trends. However, this potential is severely compromised by inconsistencies in the zonal systems used to

store these data.

This paper has explored the issues associated with designing a temporally consistent set of zones in the context of a much larger project comparing inter-regional migration in Britain and Australia in an age period-cohort framework. Such comparisons require harmonisation of the data on a number of dimensions. We outlined four possible approaches to the problem of establishing time series data on a consistent spatial basis, and pointed out a number of constraints in the context of data on internal migration. In particular, it was argued that conventional techniques for translating a data coverage from one geography to another, such as spatial interpolation, were inappropriate for migration data. In addition, the range of spatial choices was restricted because migration is not necessarily coded to the smallest spatial unit employed at the Census. Of the approaches considered, the only viable solution was the assembly of these smaller building blocks to generate designer zones with coincident outer boundaries across a number of Censuses.

It has been demonstrated that even at the level of SDs there have been substantial boundary changes in Australia over the past two decades. We have differentiated these into a number of types depending on whether they involve whole or part SLAs, and the size of the population in the area transferred. It is the changes involving the relocation of SLA boundaries that create the most significant problems for temporal continuity. Unfortunately, these are the majority. We have also demonstrated that, despite the ready availability of sophisticated GIS software, there is no simple method of automating the design of temporally consistent zones. Differences in the digital boundaries created at successive censuses fundamentally undermine the development of an automated solution.

In the face of these difficulties, the solution adopted here necessarily involved a laborious manual procedure based on visually checking overlays of SD boundaries at successive Censuses in association with published documentation to identify where boundary changes had taken place, estimation of the area and population involved and, where necessary, a subsequent search for the nearest

consistent boundary. The TSDs defined in this way represent a compromise between the competing goals of temporal consistency and maintenance of the functional integrity of the original SDs.

While alternative solutions are clearly possible, the TSD system represents a unique spatial structure that enables reliable comparisons to be made across four Censuses. The principal shortcoming of such designer solutions is that they comprise a static set of boundaries. As local government restructuring and population growth continue, it is inevitable that further changes will be made to the SD boundaries used at the 2001 and subsequent Censuses, and further inconsistencies within the TSD system will emerge. As the number of temporal observations increases, maintaining a consistent zonal system will therefore become an increasingly difficult task.

Ultimately, the solution to this problem lies either in a retreat to the strategy of freezing history, or the great leap forward to georeferencing. The former would simply require the creation by the ABS of new SLAs covering the areas involved in any intercensal changes in TSD boundaries. Temporal consistency could then be maintained on the existing boundaries by simple amalgamation. Georeferencing provides greater flexibility, but involves a substantially greater financial investment. It also calls for a comprehensive and reliable address list, a facility which is not readily available in Australia. This poses a particular problem in the context of migration because to code place of previous residence also necessitates a historical list of address records.

For the foreseeable future, then, in Australia at least, the development of temporally consistent spatial systems is likely to involve designer solutions. The TSDs described in this paper for the analysis of inter-regional migration provide one such solution, but represent the largely *ad hoc* outcome of a highly labour-intensive process. If spatially consistent zonal structures are to be developed on a rigorous basis to meet other design criteria, automated procedures will become imperative. One strategy that would materially contribute to this goal is the creation of a single master set of digital boundaries. Boundary changes could

then be recorded by simply amending the appropriate segment of the digital record, thereby eliminating the need for expensive redigitising at each successive Census. The extension of existing zone design algorithms to encompass the temporal dimension should then be a straightforward procedure.

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NOTES

- (1) The look-up tables specifying the SLAs that comprise each of the 69 TSDs are available from the National Key Centre for Social Applications of GIS at the University of Adelaide, <http://www/gisca.adelaide.edu.au/kra/pd/>

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